Keystone

Will Wingrove

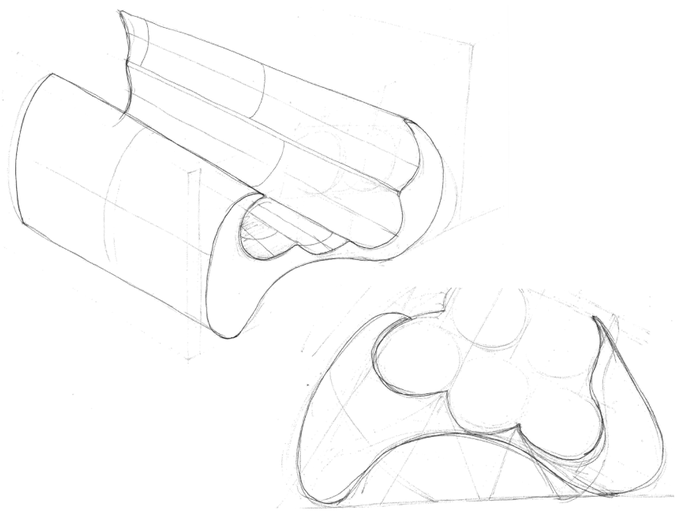
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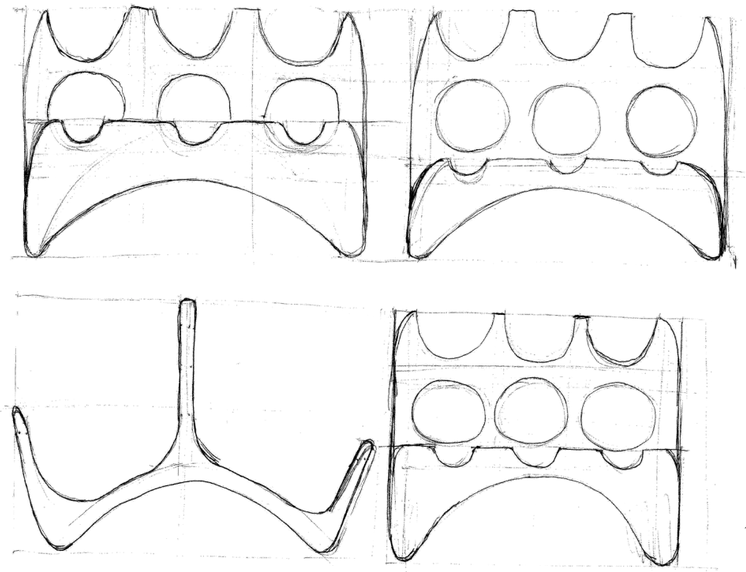
The Keystone wine rack is designed using the structural principles of arches, allowing it to securely hold up to six bottles of red wine. The weight of the bottles is evenly distributed through two arches on the front and rear, as well as slightly elongated arches on either side. This design ensures both stability and strength without the need for additional support. Beyond its functionality, Keystone was created to be both practical and visually appealing—serving as a sculptural piece that enhances its surroundings even when not in use.

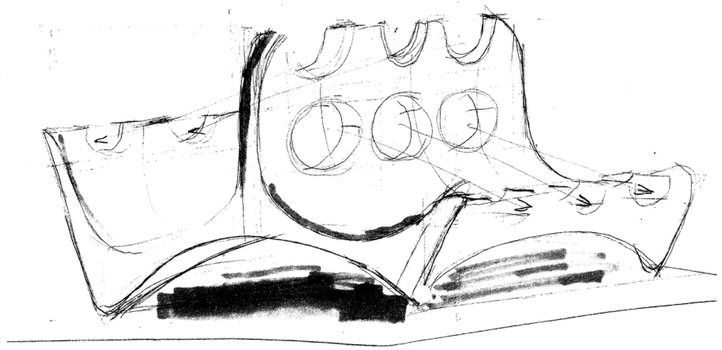


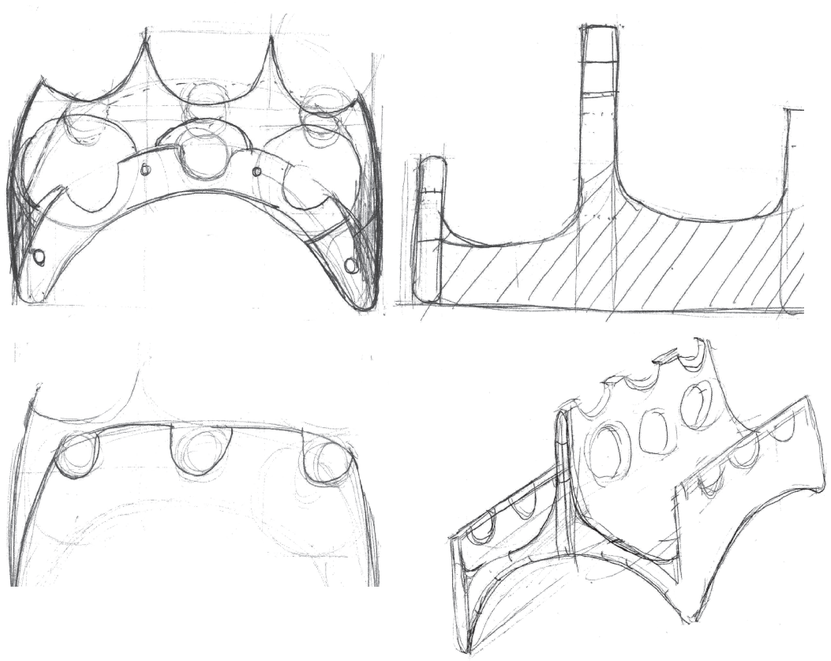
Design Process

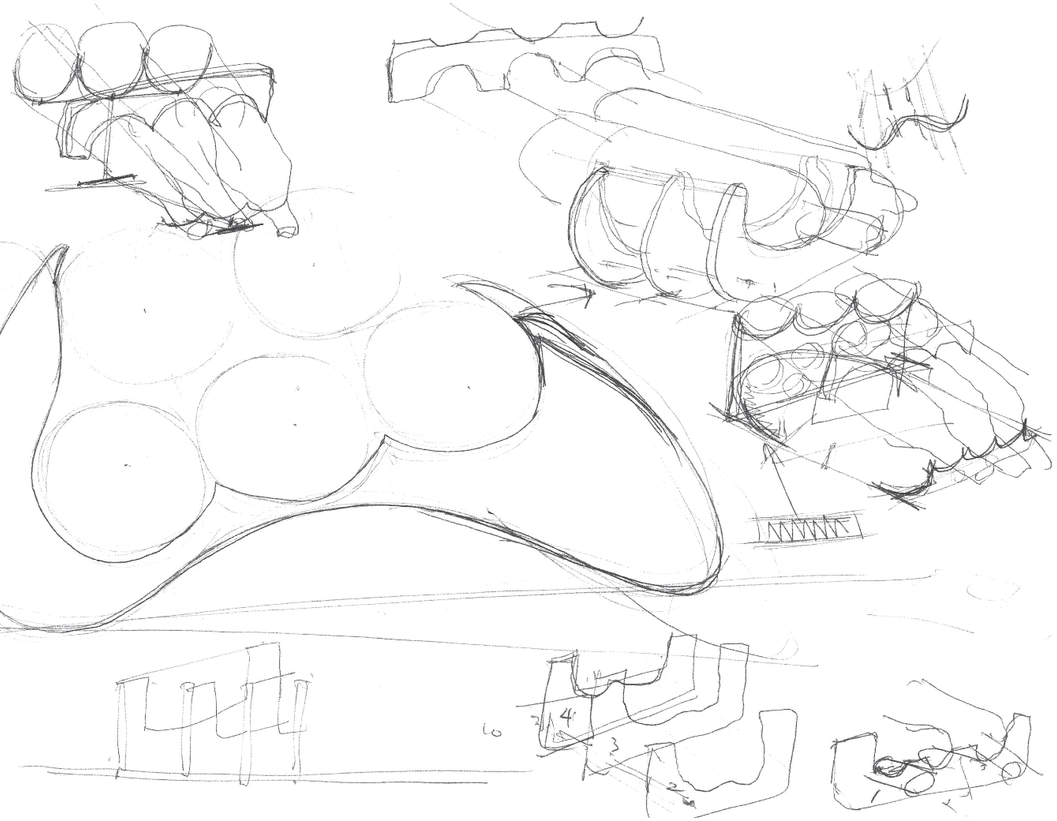
Sketches & Prototypes





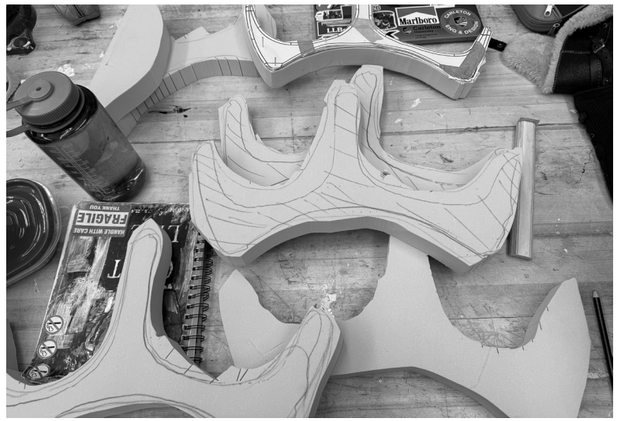






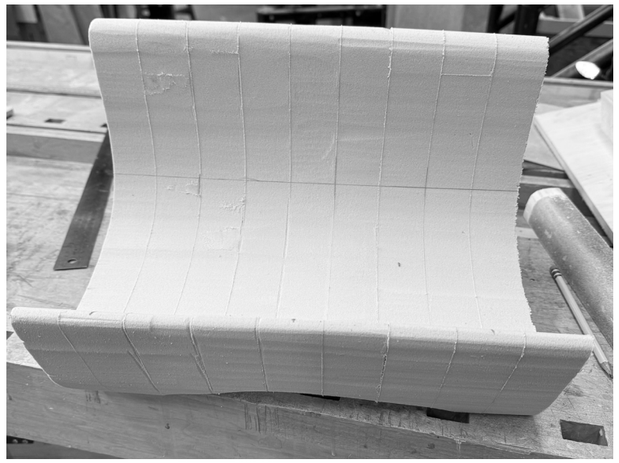
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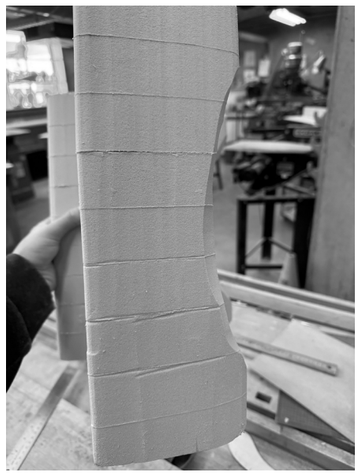


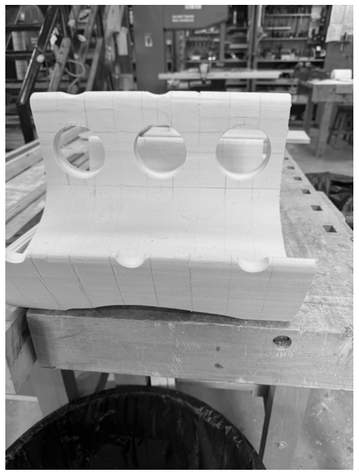


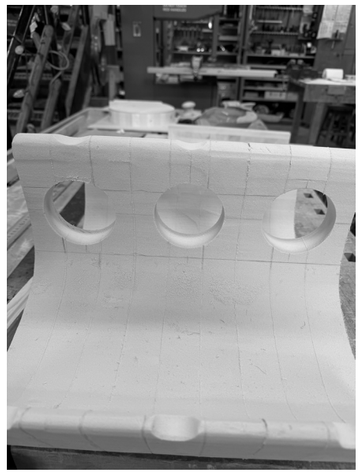


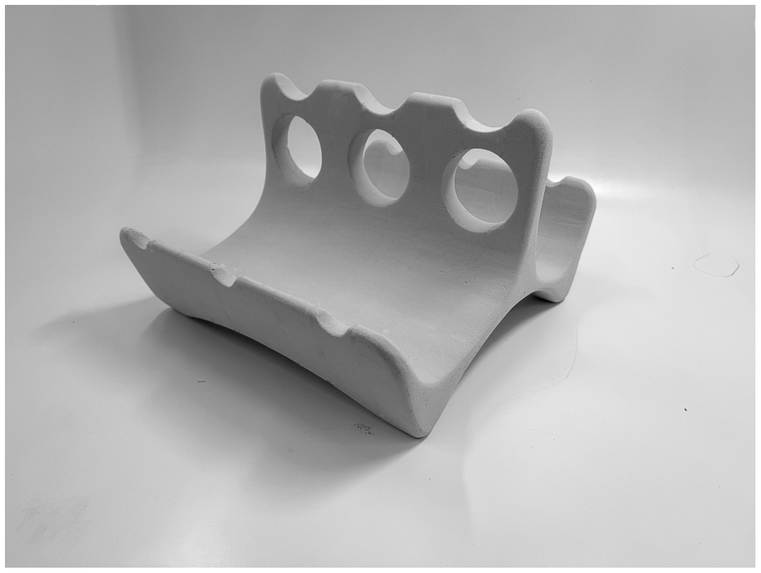






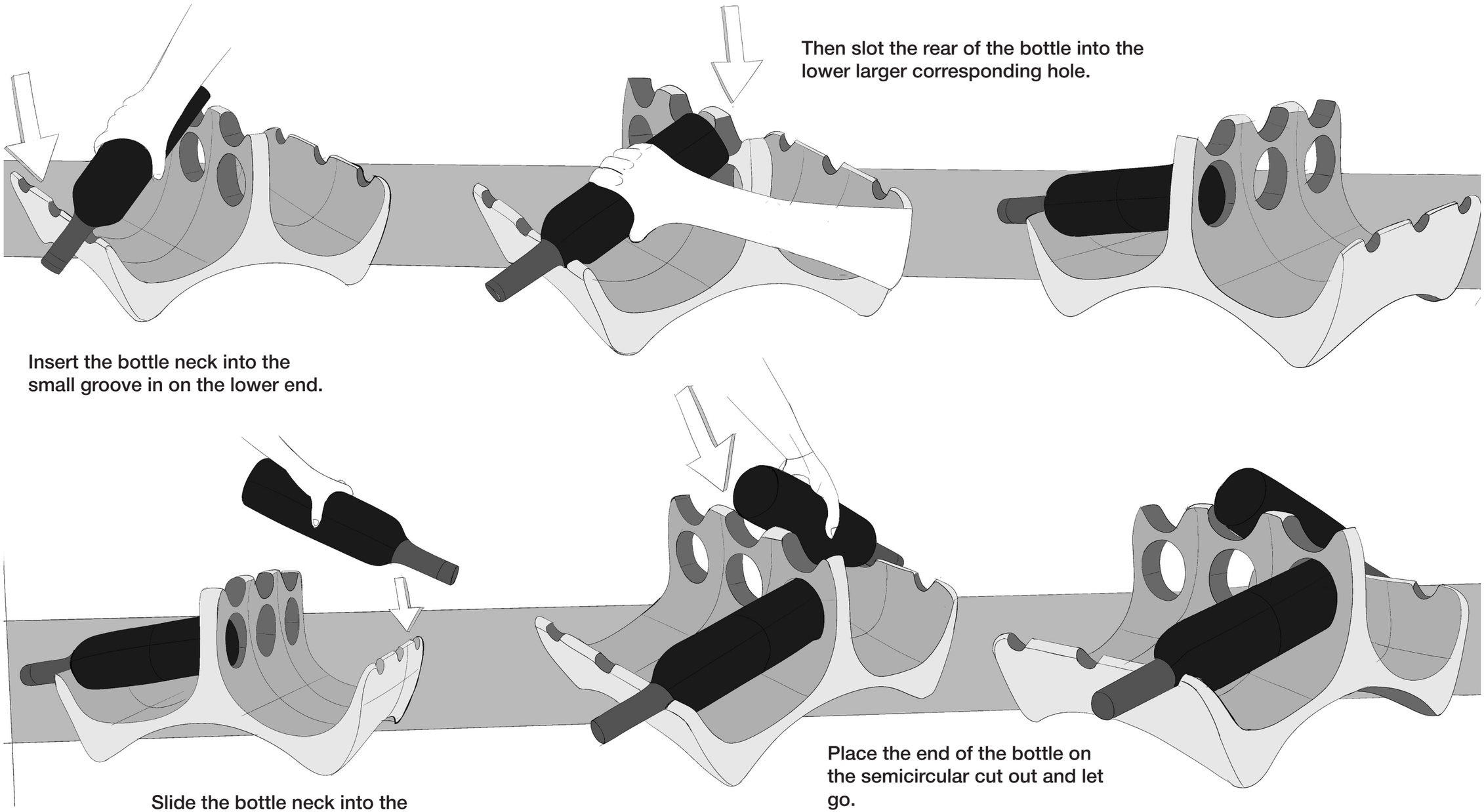






The Keystone prototype was crafted from two 3” x 5” x 1.5” sheets of high-density foam. Initially, the plan was to construct the model from laminated plywood, but due to time constraints, high-density foam was chosen instead. To achieve a final width of 13.5 inches, nine side profile layers were cut and bonded together using contact cement. The front and overall profiles were then shaped using a belt sander for precision. The three main holes through the center were cut with a 3-inch diameter hole saw and sanded to a final diameter of approximately 3.1 inches. All semicircular cutouts were meticulously shaped by hand using various sandpapers and sanding sticks. Once the final form was achieved, the surface was smoothed progressively with sandpaper grits ranging from 220 to 800, resulting in a refined finish.

Use Cycle



Insert the bottle neck into the small groove in on the lower end.

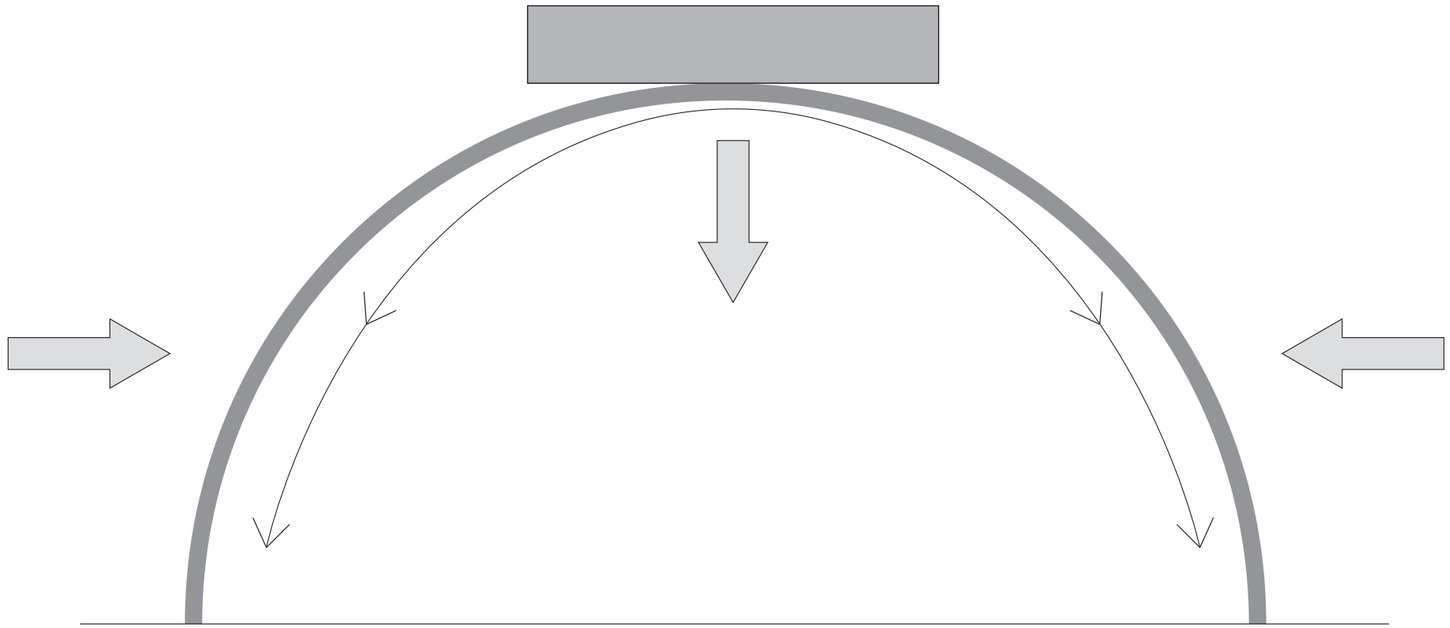
Then slot the rear of the bottle into the lower larger corresponding hole.

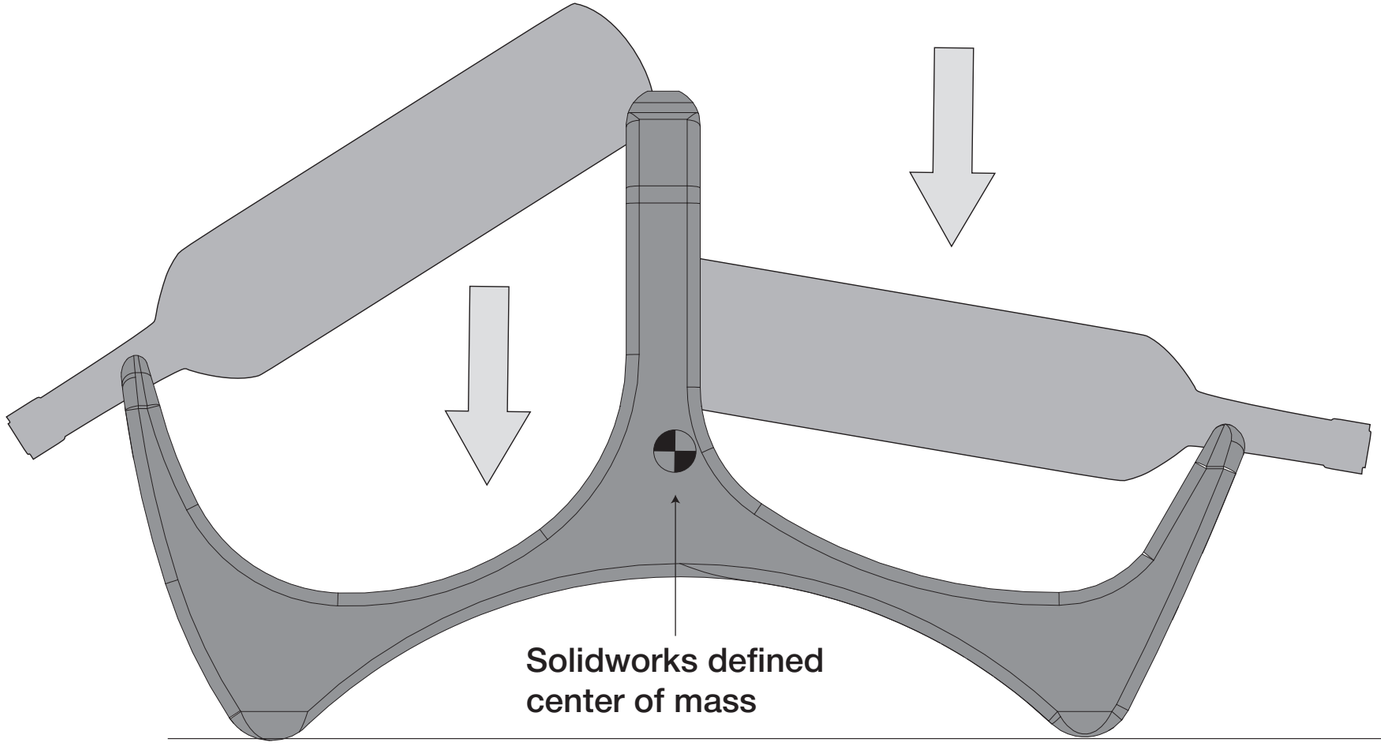
Slide the bottle neck into the groove on the higher side of the rack.

Place the end of the bottle on the semicircular cut out and let go.

Structural Principles

A basic load-bearing arch works by transferring the weight of the load above it into compression forces along its curved structure. The keystone at the top locks the arch in place, distributing the force evenly down the sides into the supporting columns or abutments. This design prevents the arch from collapsing by redirecting vertical loads into horizontal thrusts, which must be countered by strong supports. Because arches rely on compression, they can span openings without requiring additional central supports, making them efficient for bridges, doorways, and other structures.





Keystone works similarly by utilizing two load-bearing arches on both the front/back and side profiles. These arches distribute the weight of the six full-sized wine bottles, channeling the forces into compression rather than relying on an underlying support structure. The intersecting arches effectively transfer the load to the outer edges, ensuring stability and strength without additional reinforcements. This design allows Keystone to maintain structural integrity while providing a visually open and functional wine storage solution.